

PPL Lecture-6

Control Statements (Part-1)

Control Statements in C

- **Control statements** enable us to specify the flow of program control, i.e., the order in which the instructions in a program must be executed.
- They make it possible to make decisions, to perform tasks repeatedly or to jump from one section of code to another.
- There are four types of control statements in C:
 1. Decision making statements
 2. Selection statements
 3. Iteration statements
 4. Jump statements

Compound Statement or Block

- A compound statement or a block is a group of statements enclosed within a pair of curly braces { }.
- The statements inside the block are executed sequentially.

Syntax:

```
{  
    statement1;  
    statement2;  
    ⋮  
}
```

Example:

```
{  
    a=4;  
    b=2;  
    area=a*b;  
    printf("%d", area) ;  
}
```

Decision Making Statements

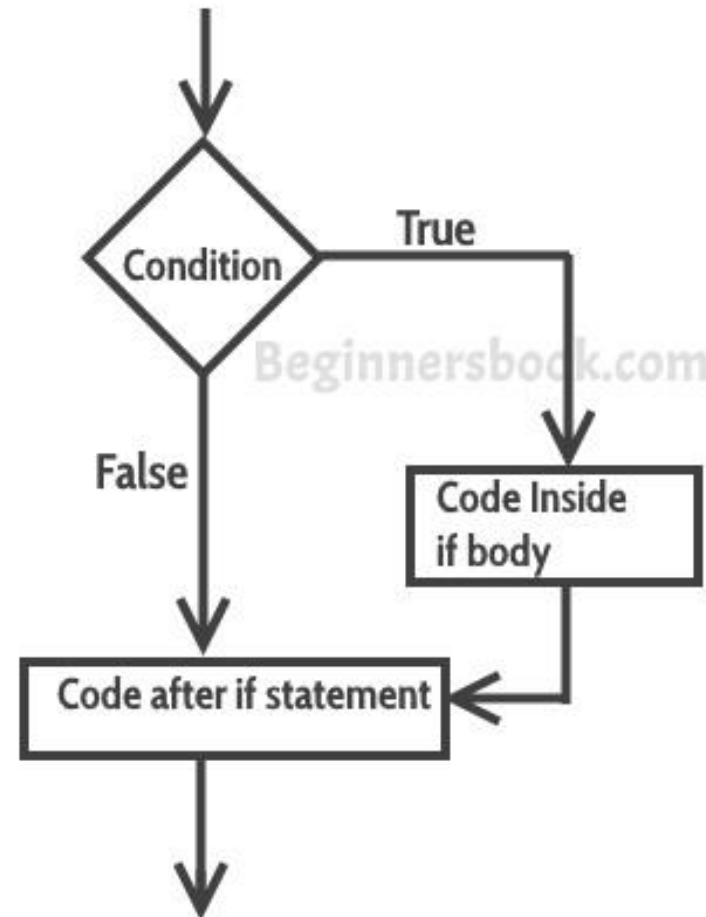
Sr.No.	Statement & Description
1	if statement An if statement consists of a boolean expression followed by one or more statements.
2	if...else statement An if statement can be followed by an optional else statement, which executes when the Boolean expression is false.
3	nested if statements You can use one if or else if statement inside another if or else if statement(s).
4	switch statement A switch statement allows a variable to be tested for equality against a list of values.

if statement

- An **if** statement consists of a *condition* followed by one or more statements.

- **Syntax:**

```
if (condition)
{
    statement(s)
}
```



Example: if statement

- Program to check whether entered number is positive or negative.

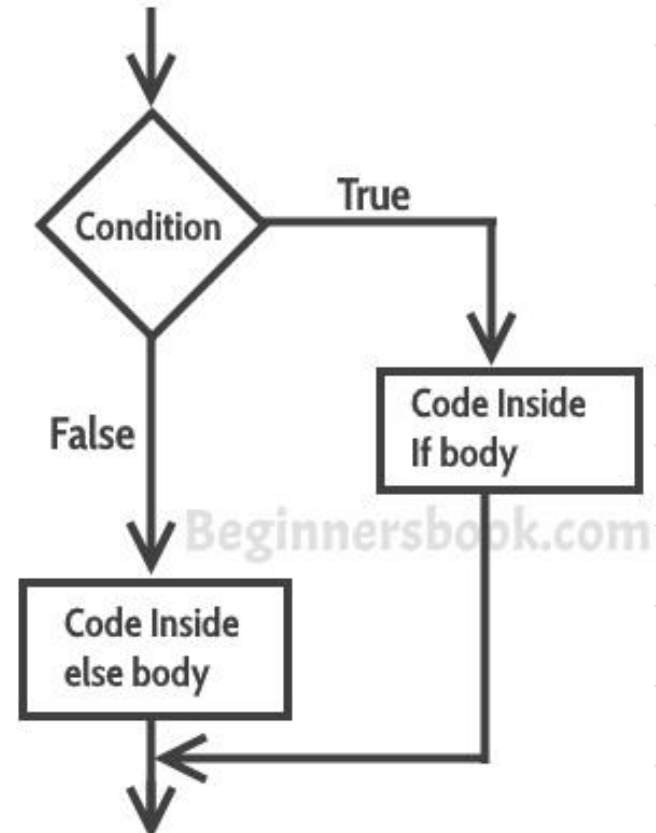
```
#include<stdio.h>
int main( )
{
    int a;
    printf("Enter a number: ");
    scanf("%d", &a);
    if(a > 0)
    {
        printf( "The number %d is positive.", a);
    }
    if(a < 0)
    {
        printf( "The number %d is negative.", a);
    }
    return 0;
}
```

if-else Statement

- The if-else statement is used to carry out a logical test and then take one of two possible actions depending on the outcome of the test (i.e., whether the outcome is true or false).

- Syntax:**

```
if (condition)
{
    statements
}
else
{
    statements
}
```



Example of if-else statement

- Program to check whether entered number is positive or negative.

```
#include<stdio.h>
int main( )
{
    int a;
    printf("\n Enter a number:");
    scanf("%d", &a);
    if(a>0)
    {
        printf( "The number %d is positive.", a);
    }
    else
    {
        printf("The number %d is negative.", a);
    }
    return 0;
}
```


Nested if and if-else Statements

```
if (condition1)
{
    // statement(s);
}
else if (condition2)
{
    //statement(s);
}
.
.
.
else if (conditionN)
{
    //statement(s);
}
else
{
    //statement(s);
}
```

Example of Nested if-else

```
if (grade== 'A')
{
    if(marks>95)
        printf("Excellent") ;
}
else
    printf ("Work hard for getting A grade");
```

Program of Nested if-else

- Program to find whether a year is leap or not

```
#include<stdio.h>
main( )
{
    int year;
    printf ("Enter year : ");
    scanf ("%d ", &year) ;
    if (year%100==0)
    {
        if (year%400==0)
        {
            printf ("Leap year\n");
        }
        else
        {
            printf ("Not leap\n");
        }
    }
}
```

Review Question

Q. Find output:

```
if( (-100 && 100) || (20 && -20) )  
    printf("%s", "Condition is true.");  
else  
    printf("%s", "Condition is false.");
```

Output: Condition is true.

Q. Which of the following are incorrect statements? If int a=10.

- 1) `if( a==10 ) printf("IncludeHelp");`
- 2) `if( 10==a ) printf("IncludeHelp");`
- 3) `if( a=10 ) printf("IncludeHelp");`
- 4) `if( 10=a ) printf("IncludeHelp");`

Review Question

Q. Find output:

```
int a=20, b=3;
```

```
if(a<10)
```

```
    a=a-5;
```

```
    b=b+5;
```

```
printf("%d %d\n",a,b);
```

Output: 20, 8

```
int i=1, j =9;
```

```
if(i>=5 && j<5);
```

```
    i=j+2;
```

```
printf("%d\n",i);
```

Output: 11

Review Question

Q. Find output:

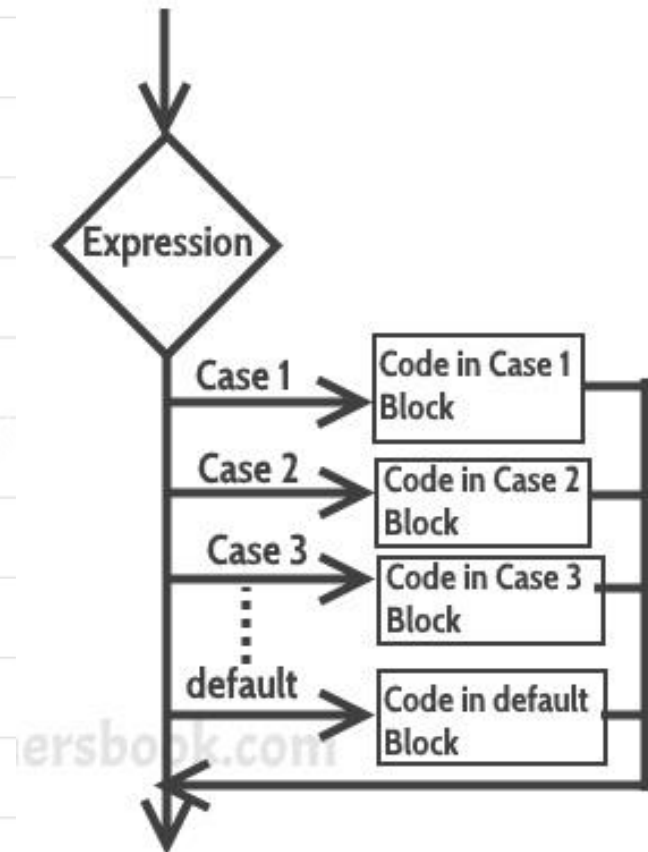
```
int a=0, b=0;
if ( !a )
{
    b = !a;
    if (b)
        a = !b;
}
printf(“%d , %d”,a,b);
```

Output: 0 , 1

Selection Statement: the switch-case Statement

- A switch statement is used for **multiple way selections** that will branch into different code segments based on the value of a variable or expression.
- This expression or variable must be of integer data type.
- **Syntax:**

```
switch (expression)
{
    case value1:
        code segment1;
        break;
    case value2:
        code segment2;
        break;
    .
    .
    .
    case valueN:
        code segmentN;
        break;
    default:
        default code segment;
}
```



Example: switch statement

```
int choice;
printf ("Enter your choice ") ;
scanf("%d",&choice);

switch(choice)
{
    case 1:
        printf("First\n");
        break;
    case 2:
        printf("Second\n");
        break;
    case 3:
        printf("Third\n");
        break;
    default:
        printf("Wrong choice");
}
```


Review Question

Q. Find output:

```
int p = 800, q, r;  
if (p >= 700)  
    q = 600;  
r = 500;  
printf("%d %d\n", q, r);
```

Output: 600 500

Q. Find output:

```
int x=22;  
if(x=10)  
    printf("TRUE");  
else  
    printf("FALSE");
```

Output: TRUE

Review Question

Q. Find output:

```
int num=2;
switch(num+2)
{
    case 1:
        printf("Case1: Value is: %d", num);
    case 2:
        printf("Case1: Value is: %d", num);
    case 3:
        printf("Case1: Value is: %d", num);
    default:
        printf("Default: Value is: %d", num);
}
```

Output: Default: Value is: 2

Review Question

Q. Find output:

```
int i=2;
switch (i)
{
    case 1:
        printf("Case1 ");
        break;
    case 2:
        printf("Case2 ");
    case 3:
        printf("Case3 ");
        break;
    case 4:
        printf("Case4 ");
    default:
        printf("Default ");
}
```

Output: Case2 Case3

Review Question

Q. Find output:

```
int a=2;
```

```
int b=a;
```

Output: Case-a

Exit from switch

```
switch(b)
```

```
{
```

```
    case a:
```

```
        printf("Case-a\n");    break;
```

```
    case 3:
```

```
        printf("Case-3\n");    break;
```

```
    default:
```

```
        printf("No option\n"); break;
```

```
}
```

```
printf("Exit from switch");
```